

Recommender Systems

Harnessing the Power of Personalization

Michael Hahsler

Engineering Management, Information, and Systems (EMIS)
Intelligent Data Analysis Lab (IDA@SMU)
Bobby B. Lyle School of Engineering, Southern Methodist University

Curricular Recommender System Working Group
February 17, 2017



SMU | BOBBY B. LYLE
SCHOOL OF ENGINEERING

Mission: At IDA we create novel techniques inspired by knowledge discovery, data mining, machine learning, artificial intelligence and statistical analysis to work with data from various sources. We currently focus on:

- Order modeling for massive data streams with applications in meteorology (hurricane intensity prediction) and personalized medicine (efficient classification and analysis of metagenomic data)
- Visual analytics using optimized reordering
- Simulation data analytics
- Recommender systems

Team: 3 faculty, 7 students, 2 collaborators

Director: M. Hahsler <mhahsler@lyle.smu.edu>

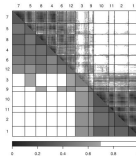
Supported by



National Human
Genome Research
Institute



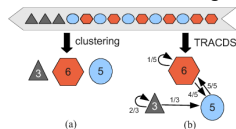
Visual Analytics



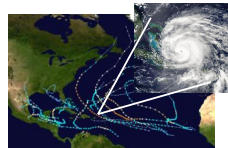
Simulation Data Analytics



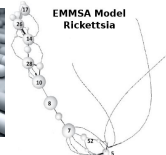
Order Modeling Data Stream Clustering



Meteorology



Personalized Medicine



Reproducible Research Using R



R has been consistently voted one of the most important tools for data mining and analytics, and being able to use R is one of the highest paying analytics skills.

Our team has developed and maintains several popular R packages:

Association Rule Mining

- **arules**: Mining association rules and frequent itemsets.
- **arulesViz**: Visualizing association rules based on package arules.
- **arulesSequences**: Mine frequent sequences.

Combinatorial Optimization

- **seriation**: Seriation/sequencing techniques to reorder matrices and dendrograms.
- **TSP**: Infrastructure and algorithms for the traveling salesperson problem.
- **DBSCAN**: Several density-based algorithms for spatial data.
- **QAP**: Heuristics for the Quadratic Assignment Problem (QAP).

Data Stream Mining

- **stream**: Infrastructure for data stream mining.

Recommender Systems

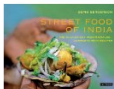
- **recommenderlab**: Infrastructure to test and develop recommender algorithms.

<http://michael.hahsler.net/#Software>



Kristina, Welcome to Your Amazon.com

Today's Recommendations For You

Here's a daily sample of items recommended for you. Click here to [see all recommendations](#).

[Street Food of India: The 50...
\(Hardcover\) by Septh
Bergerson](#)

★★★★☆ (4) \$19.17

[Fix this recommendation](#)

[Lavazza Tierra! 100% Arabica
Whole Bean Espresso...](#)

★★★★☆ (38) \$34.41

[Fix this recommendation](#)

[Entourage: The Complete
Fou... DVD ~ Adrian Grenier](#)

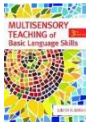
★★★★☆ (44) \$16.49

[Fix this recommendation](#)

New For You®



[The Race \(Isaac Bell\)
Clive Cussler, Justin Scott
Hardcover](#)

~~\$27.95~~ \$14.97[Fix this recommendation](#)

[Multisensory Teaching of
Basic...
Judith R. Birsh, Sally E.
Shaywitz
Hardcover](#)

~~\$79.95~~ \$44.99[Fix this recommendation](#)

[Kill Shot \(Mitch Rapp\)
Vince Flynn
Hardcover](#)

~~\$27.99~~ \$16.62[Fix this recommendation](#)

[Limitless \(Unrated
Extended Cut\)
Bradley Cooper, Anna
Friel, Abbie...
DVD](#)

~~\$29.99~~ \$15.19[Fix this recommendation](#)

Suggestions (1141)

Suggestions by Genre ▾

Rate Movies

Rate Genres

Movies You've Rated (262)

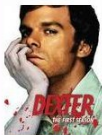
Movies You'll Love

Suggestions based on your ratings

 You have 11.41
 Suggestions
 from 262 ratings.

New Suggestions for you

Based on your recent ratings



Play + All



Not Interested

 Dexter
 1 (4)
 Bec
 enjo
 Lost
 Battl
 Gala
 Seas
 Rom

The Fugitive (1993)

Wrongfully convicted of murdering his wife, Dr. Richard Kimble (Harrison Ford) escapes custody after a ferocious train accident (one of the most thrilling wrecks ever filmed). While Kimble tries to find the true murderer, gung-ho U.S. Marshal Samuel Gerard (Tommy Lee Jones, in an Oscar-winning performance) is hot on Kimble's trail, pulling out all stops to put him back behind bars.

Starring: Harrison Ford, Tommy Lee Jones

Director: Andrew Davis

Genre: Action & Adventure

MPAA: PG-13


4.7 Our best guess for Michael



4.1 Customer Average

Recommended based on 8 ratings

nos:

e

su

ther

ther,

nder:



Add



Not Interested

The Fugitive

Because you enjoyed:

[Patriot Games](#)
[Indiana Jones and the Last Crusade](#)
[Die Hard](#)
[Die Hard](#)


SCI-FI &



Incredi

See all 26 >



Spacehunter



RoboCop:

Can I recommend anything else?

<http://www.kdnuggets.com/2015/10/big-data-recommendation-systems-change-lives.html/2>

Table of Contents

- 1 Definition: Recommender Systems
- 2 Content-based Approach
- 3 Collaborative Filtering (CF)
 - Memory-based CF
 - Model-based CF
 - Strategies for the Cold Start Problem
 - Hybrid Recommender Systems
- 4 Implementation
- 5 Curricular Recommender Systems

Recommender Systems

Original Definition

Recommender systems apply statistical and knowledge discovery techniques to the problem of making product recommendations.

Sarwar *et al.* (2000)

Advantages of recommender systems (Schafer *et al.*, 2001):

- Improve conversion rate: Help customers find a product she/he wants to buy.
- Cross-selling: Suggest additional products.
- Up-selling: Suggest premium products.
- Improve customer loyalty: Create a value-added relationship.

A More General View of Recommender Systems

A recommender system is a **fully automatic system** to provide (near) **personalized decision support** given **limited information** while optimizing a set of potentially conflicting **objective functions**.

A More General View of Recommender Systems

A recommender system is a **fully automatic system** to provide (near) **personalized decision support** given **limited information** while optimizing a set of potentially conflicting **objective functions**.

Important aspects:

- Personalization
- Available information
- Incentive structure
- Trust
- Quality of recommendations
- Speed

Common Approaches

- **Content-based filtering:** Consumer preferences for product attributes.
- **Collaborative filtering:** Mimics word-of-mouth based on analysis of rating/usage/sales data from many users.

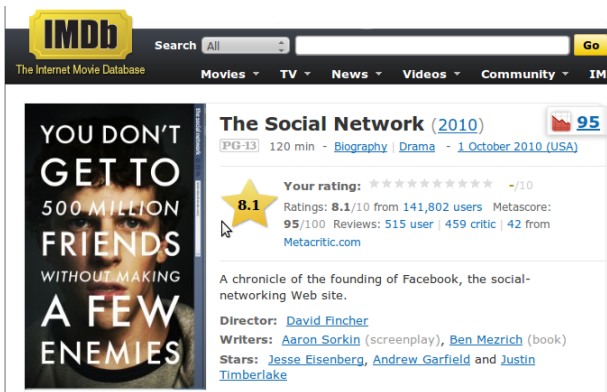
(Ansari *et al.*, 2000)

- **Hybrid recommender systems:** Incorporate content, collaborative and expert information.

Table of Contents

- 1 Definition: Recommender Systems
- 2 Content-based Approach**
- 3 Collaborative Filtering (CF)
 - Memory-based CF
 - Model-based CF
 - Strategies for the Cold Start Problem
 - Hybrid Recommender Systems
- 4 Implementation
- 5 Curricular Recommender Systems

Content-based Approach



IMDb
The Internet Movie Database

Search

Movies ▾ **TV** ▾ **News** ▾ **Videos** ▾ **Community** ▾ **IMDb**

YOU DON'T GET TO 500 MILLION FRIENDS WITHOUT MAKING A FEW ENEMIES

The Social Network (2010) 

PG-13 120 min - [Biography](#) | [Drama](#) - [1 October 2010 \(USA\)](#)

Your rating: ★★★★★★★★ -/10
Ratings: **8.1**/10 from [141,802 users](#) Metascore:
95/100 Reviews: [515 user](#) | [459 critic](#) | [42 from Metacritic.com](#)

A chronicle of the founding of Facebook, the social-networking Web site.

Director: [David Fincher](#)
Writers: [Aaron Sorkin](#) (screenplay), [Ben Mezrich](#) (book)
Stars: [Jesse Eisenberg](#), [Andrew Garfield](#) and [Justin Timberlake](#)

- 1 Analyze the objects (documents, video, music, etc.) and extract attributes/features (e.g., words, phrases, actors, genre).
- 2 Recommend objects with similar attributes to an object the user likes.



Lady Gaga

[Just Dance \(Remix Single\)](#)

Just Dance (Redone Remix F. Kardinal Offishall)

Play Sample

PANDORA®

Features Of This Track

electronica roots
trip hop roots
r&b influences
funk influences
beats made for dancing
unsyncopated ensemble rhythms
straight drum beats
a female vocal
clear pronunciation
a rhythmic intro
use of modal harmonies
the use of chordal patterning
melodic part writing
use of strings
subtle use of arpeggiated synths
affected synths

Create A Station

Bookmark This Track

Buy on iTunes

Buy CD From Amazon

Buy From Amazon MP3

“The **Music Genome Project** is an effort to capture the essence of music at the fundamental level using almost 400 attributes to describe songs and a complex mathematical algorithm to organize them.”

http://en.wikipedia.org/wiki/Music_Genome_Project

An issue with content based filtering?

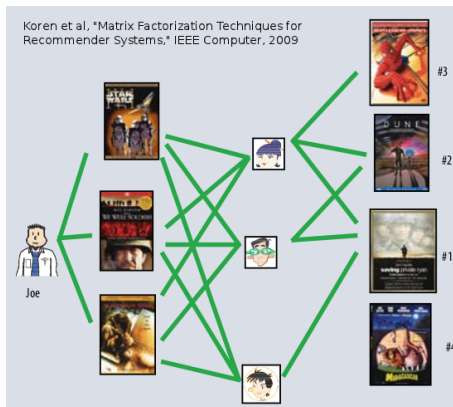
An issue with content based filtering?

Missing diversity!

Table of Contents

- 1 Definition: Recommender Systems
- 2 Content-based Approach
- 3 Collaborative Filtering (CF)
 - Memory-based CF
 - Model-based CF
 - Strategies for the Cold Start Problem
 - Hybrid Recommender Systems
- 4 Implementation
- 5 Curricular Recommender Systems

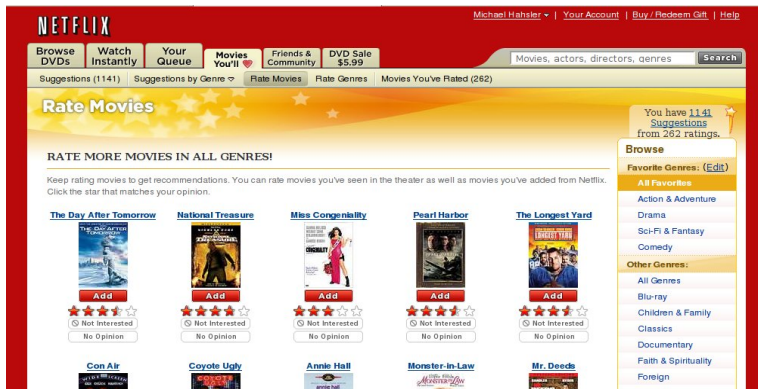
Collaborative Filtering (CF)



Make automatic predictions (filtering) about the interests of a user by collecting preferences or taste information from many other users (collaboration).

Assumption: those who agreed in the past tend to agree again in the future.

Data Collection for CF

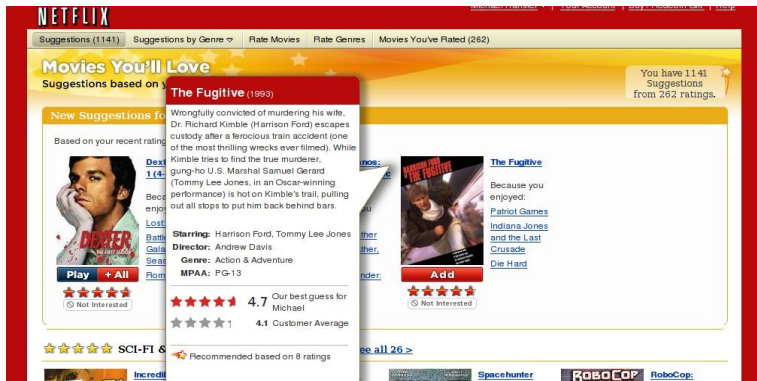


• Data sources:

- ▶ **Explicit:** ask the user for ratings, rankings, list of favorites, etc.
- ▶ **Observed behavior (Implicit):** clicks, page impressions, purchase, uses, downloads, posts, tweets, etc.

What is the incentive structure?

Output of a Recommender System

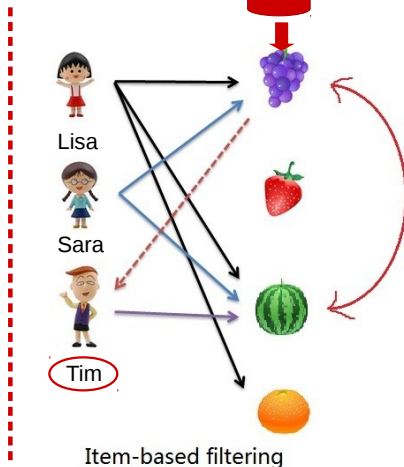
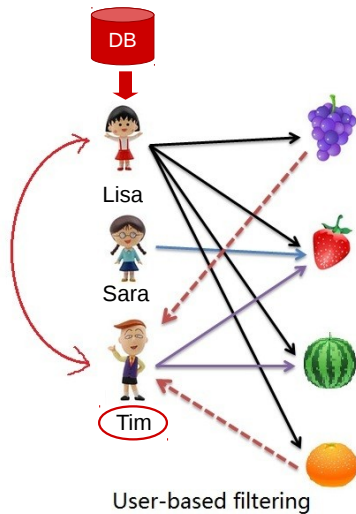


- Predicted rating of unrated movies (Breese *et al.*, 1998)
- A top- N list of unrated (unknown) movies ordered by predicted rating/score (Deshpande and Karypis, 2004)

Table of Contents

- 1 Definition: Recommender Systems
- 2 Content-based Approach
- 3 Collaborative Filtering (CF)**
 - Memory-based CF
 - Model-based CF
 - Strategies for the Cold Start Problem
 - Hybrid Recommender Systems
- 4 Implementation
- 5 Curricular Recommender Systems

User-based vs. Item-based CF



Source: <http://cuihelei.blogspot.com/2012/09/the-difference-among-three.html>

Table of Contents

- 1 Definition: Recommender Systems
- 2 Content-based Approach
- 3 Collaborative Filtering (CF)**
 - Memory-based CF
 - Model-based CF**
 - Strategies for the Cold Start Problem
 - Hybrid Recommender Systems
- 4 Implementation
- 5 Curricular Recommender Systems

Different Model-based CF Techniques

There are many techniques:

- **Cluster users** (i.e., customer segmentation) and then recommend items the users in the cluster closest to the active user like.
- Mine **association rules** (if-then rules) and then use the rules to recommend items.
- Define a null-model (a stochastic process which models usage of independent items) and then find **significant deviation from the null-model**.
- **Learning to rank**: Logistic regression, neural networks (deep learning) and many other machine learning methods.
- Learn a **latent factor model** from the data and then use the discovered factors to find items with high expected ratings.
- Matrix completion using **matrix factorization**.

Table of Contents

- 1 Definition: Recommender Systems
- 2 Content-based Approach
- 3 Collaborative Filtering (CF)**
 - Memory-based CF
 - Model-based CF
 - Strategies for the Cold Start Problem**
 - Hybrid Recommender Systems
- 4 Implementation
- 5 Curricular Recommender Systems

Cold Start Problem

What do we recommend to new users for whom we have no ratings yet?

Cold Start Problem

What do we recommend to new users for whom we have no ratings yet?

- Recommend popular items
- Have some start-up questions (e.g., "What are your 10 favorite movies?")
- Obtain/purchase personal information

Cold Start Problem

What do we recommend to new users for whom we have no ratings yet?

- Recommend popular items
- Have some start-up questions (e.g., "What are your 10 favorite movies?")
- Obtain/purchase personal information

What do we do with new items?

Cold Start Problem

What do we recommend to new users for whom we have no ratings yet?

- Recommend popular items
- Have some start-up questions (e.g., "What are your 10 favorite movies?")
- Obtain/purchase personal information

What do we do with new items?

- Content-based filtering techniques.
- Use expert/domain knowledge.
- Pay a focus group to rate new items.

Table of Contents

- 1 Definition: Recommender Systems
- 2 Content-based Approach
- 3 Collaborative Filtering (CF)
 - Memory-based CF
 - Model-based CF
 - Strategies for the Cold Start Problem
 - Hybrid Recommender Systems
- 4 Implementation
- 5 Curricular Recommender Systems

Hybrid Recommender Systems

Combines recommendations using different strategies or data. For example:

- Collaborative filtering
- Content-based filtering
- Expert knowledge
- Popular items
- Re-recommending items the user liked (e.g., for songs or consumables).
- Add random or new items to increase variety.
- Consider business objectives.

Table of Contents

- 1 Definition: Recommender Systems
- 2 Content-based Approach
- 3 Collaborative Filtering (CF)
 - Memory-based CF
 - Model-based CF
 - Strategies for the Cold Start Problem
 - Hybrid Recommender Systems
- 4 Implementation**
- 5 Curricular Recommender Systems

Open-Source Implementations

- **Apache Mahout:** ML library including collaborative filtering (Java)
- **C/Matlab Toolkit for Collaborative Filtering** (C/Matlab)
- **Cofi:** Collaborative Filtering Library (Java)
- **Crab:** Components for recommender systems (Python)
- **easyrec:** Recommender for Web pages (Java)
- **LensKit:** CF algorithms from GroupLens Research (Java)
- **MyMediaLite:** Recommender system algorithms. (C#/Mono)
- **RACOFI:** A rule-applying collaborative filtering system
- **Rating-based item-to-item recommender system** (PHP/SQL)
- **recommenderlab:** Infrastructure to test and develop recommender algorithms (R)

See <http://michael.hahsler.net/research/recommender/> for URLs.

Table of Contents

- 1 Definition: Recommender Systems
- 2 Content-based Approach
- 3 Collaborative Filtering (CF)
 - Memory-based CF
 - Model-based CF
 - Strategies for the Cold Start Problem
 - Hybrid Recommender Systems
- 4 Implementation
- 5 Curricular Recommender Systems

Degree Compass

The screenshot displays the Degree Compass interface within a Firefox browser. On the left, a sidebar shows details for 'BIOL 1010: Principles of Life', including its course description, a note about prerequisites, and class section information for Spring Semester 2011. The main content area, titled 'Courses To Consider', lists recommended courses with star ratings and 'View Sections' links. A filter box at the bottom allows users to refine suggestions based on previous successful progress.

BIOL 1010: Principles of Life

Course Description: A course for non-science majors. Topics covered include scientific methodology, the nature of living organisms, cell structure and function, cell chemistry and division, nature of heredity and gene action, the theory of evolution and principles of ecology. BIOL 1010 will not serve as a prerequisite of upper level biology courses.

Note: To add any of the sections below to your class schedule, return to the main OneStop window, click on the "Web Self Service" tab, then "Student", then "Registration", then "Add or Drop Classes". You'll also want to make note of the CRN for the course you wish to register for as this will make finding the class in the registration system easier.

Spring Semester 2011

Class Section: 01

Class CRN: 1135
Instructor: Finley, Mack
Credit Hours: 3
Time: 08:00 am - 08:55 am
Days: MWTF
Campus: Austin Peay SU, Main Campus
Location: Sundquist Science Complex E106A
Instructional Method: Conventional Methodology
Start Date: 13-JAN-11
End Date: 06-MAY-11
Capacity: 99
Seats Open: 98
Seats Filled: 1

Courses To Consider

Courses You Should Consider:

- BIOL1010 - Principles of Life (5 stars) [View Sections](#)
- BIOL1011 - Principles of Life Lab (4 stars) [View Sections](#)
- BIOL1010 - Principles of Life (4 stars) [View Sections](#)
- GEO1041 - Physical Geology Lab (4 stars) [View Sections](#)
- BIOL2011 - Human Anat and Phys Lab (4 stars) [View Sections](#)
- GEO1040 - Physical Geology (4 stars) [View Sections](#)

[See more suggestions...](#)

Filter: MATH, ENGL, etc.

These suggestions are courses in which other students similar to you have made successful progress in your program of study.
You should always consult your advisor when planning your schedule.

My Courses

AP Austin Peay Online
Wherever you go, there we are.

APOnline
Use this link to access your WL, WL2, etc. APOnline course sections.

Notes: If you are planning to browse your course for longer than 30 minutes (or to take a quiz or post a forum topic), please use the direct login, <http://learn.apstate.edu> to avoid OneStop timing out your online course session.

RDDP
Click here to access your RDD, RSL, etc. courses. (Requires log-in)
RDDP Login Instructions
Login instructions are located on the RDDP Online login page.

“Predict in which course a student will do well based on past performance.”

There are lots of questions!

- What **“items”** do we want to recommend? Programs, courses, second majors/minors?
- **Content-based filtering:** What features do items (e.g., courses) have? Learning outcomes, prerequisites, tags?
- **Collaborative filtering:** How do students rate courses? Stated level of interest in course or topic, grades in other courses?
- **Expert knowledge:** Is this encoded in degree plans created by advisers for previous students?
- **Enforce rules in the curriculum:** This is hard since it is not the typical recommender system setting.
- How do we deal with **new items** (courses/programs)?
- **Balance demand** for courses/programs?

Thank you!

This presentation can be downloaded from
<http://michael.hahsler.net/> (under publications/talks)

For questions, please contact the author at mhahsler@lyle.smu.edu

recommenderlab is available in R from CRAN.

An introduction can be found at <https://cran.r-project.org/web/packages/recommenderlab/vignettes/recommenderlab.pdf>

References I

- Asim Ansari, Skander Essegaier, and Rajeev Kohli. Internet recommendation systems. *Journal of Marketing Research*, 37:363–375, 2000.
- John S. Breese, David Heckerman, and Carl Kadie. Empirical analysis of predictive algorithms for collaborative filtering. In *Proceedings of the Fourteenth Annual Conference on Uncertainty in Artificial Intelligence*, pages 43–52, 1998.
- Mukund Deshpande and George Karypis. Item-based top-n recommendation algorithms. *ACM Transactions on Information Systems*, 22(1):143–177, 2004.
- Badrul Sarwar, George Karypis, Joseph Konstan, and John Riedl. Analysis of recommendation algorithms for e-commerce. In *EC '00: Proceedings of the 2nd ACM conference on Electronic commerce*, pages 158–167. ACM, 2000.
- J. Ben Schafer, Joseph A. Konstan, and John Riedl. E-commerce recommendation applications. *Data Mining and Knowledge Discovery*, 5(1/2):115–153, 2001.